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STATUS OF VITAMIN A IN IRON DEFICIENT ANEMIC WOMEN OF REPRODUCTIVE AGE GROUP

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ABSTRACT
The prevailing hypothesis of association between vitamin A and iron metabolism in the etiology of iron deficiency anemia was explored in a study conducted on 100 Pakistani women (ages 13 to 45 years). Estimations were carried out for hemoglobin, hematocrit, RBC indices, serum levels of retinol, iron, transferrin & ferritin and TIBC. It was concluded that vitamin A status of Pakistani anemic women is adequate and does not appear to be a predisposing factor in the etiology of iron deficiency anemia.

INTRODUCTION

Scattered reports all over the world suggest that anemia can be a consequence of vitamin A deficiency. A conservative estimate indicates the prevalence of anemia is at around 36% or 1200 million people in the developing world (WHO/FAO). Nutritional anemia and vitamin A deficiency have been shown to be two of the major nutritional problems not only in the developing countries, but also in the highly developed nations (Report of WHO/FAO Expert Consultation Group, 1988).

An inadequate intake of vitamin A appears to be the under lying cause of low plasma retinol levels, while nutritional anemia has been traditionally associated with deficiency of iron, folate, vitamin B12 and protein.

A direct association between vitamin A deficiency and Iron deficiency anemia was first investigated in 1922 (Findlay & Mackenzie) and since then numerous investigators have repeatedly shown a positive correlation (Wolback & Howe, 1925; Koessler *et al.*, 1926; Sure *et al.*, 1933; Frank M. *et al.*, 1934; Wagner K.H., 1940; O'Tool *et al.*, 1977). However, these observations have also been contradicted by other reports of apparent polycythemia that probably resulted from hemoconcentration that sometimes occur in hypervitaminosis A (Amine *et al.*, 1970).

The prevailing hypothesis explaining the association between vitamin A and iron metabolism, are:

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1. Vitamin A influences the differentiation of the red cell.
2. It mobilizes iron from its stores in reticulo-endothelial tissues to the bone marrow, for utilization (O'Tool 1974; Hodges *et al.*, 1977; Mohanram 1977; Muhilal 1988).
3. Vitamin A deficiency increases the susceptibility to infectious diseases and this causes impaired hematopoiesis.

Thus in vitamin A deficiency (a) anemia develops which is refractory to medicinal iron, but responds to vitamin A, and (b) vitamin A supplementation elevates levels of iron in storage depots especially the liver(Meija *et al.*, 1977). However in severe vitamin A deficiency, there is hypovolemia and hemoconcentration and this can mask the anemia and even result in an apparent polycythemia (Amine et al., 1970).

In spite of numerous international studies on the subject, very little work has previously been done in Pakistan.

Plasma vitamin A levels were estimated in 1036 healthy subjects of various ages and socio-economic groups indicated that only children under the age of 15 years are at risk in Karachi (Khushnaseeb et al., 1987).

More recently, a study on anemic children from low income groups in Karachi, showed that periodic massive dose of vitamin A although, was instrumental in improving vitamin A status of these children, but failed to show an improvement in the iron status.

The present study was designed keeping in mind that:

1. Nutritional anemia is a major health problem in Asian subcontinent.
2. Asian population has been neglected till date.
3. Genetic, racial, climatic, dietary, and geographic variation demands studies to be carried out in the region.
4. Vitamin A requirements and status may or may not be the same as that of western population.
5. Response to supplementation and subsequent correction of vitamin A status in this population may or may not be the same in the western world.

The main objective of the present study, therefore, was to see:

a) Weather vitamin A status of the most vulnerable population (menstruating females) was adequate or not.
b) Weather anemic women were also vitamin A deficient or not.
c) Feasibility of improving vitamin A status of vitamin A deficient anemic women in order to improve general health of the female population.

MATERIALS AND METHODS

80 Anemic women (13-45 y) were randomly selected from urban slums of Karachi, belonging to low socio-economic strata. 20 healthy women with adequate hemoglobin status of the same age group and social strata, were also included in the study as controls. All 100 women underwent initial hemoglobin estimation on the spot for selection purpose, by Sahli's hemoglobin meter. Once selected, 10 cc of blood was collected from each subject. Each sample was then tested for the following parameters.

I. Hemoglobin
II. Hematocrit
III. Serum retinol
IV. Serum iron
V. Serum transferrin
VI. Serum ferritin
VII. RBC count

RESULTS

Normal values for the parameters investigated in the study are given in Table 1. It was observed in the study that 5 out of 80 women in the study group were vitamin A deficient (highly insignificant) although all of them were anemic according to world standards (hemoglobin < 10 gms. %), Table 2. None of the 20 women in the control group were found to be vitamin A deficient.

DISCUSSION

Nutritional anemia has been a subject of considerable interest for many decades now. It may result from nutritional deficiencies of a variety of vitamins and trace minerals.

Iron, being an integral component of hemoglobin, Myoglobin, and the iron containing enzymes essential for intermediary metabolism, is considered as the main culprit in its development.

Vitamin deficiencies that have been implicated as causes of anemia in humans are folic acids, vitamin B_{12}, A, C and other members of the B group. Vitamin E and copper are also recognized as essential for erythropoiesis.

Previously some related work has been done in Pakistan and abroad.

Mushtaq and Jalil (1985) conducted a study to determine the feasibility of fortification best suited to Pakistan, in order to overcome the iron deficiency anemia. In this study they supplemented individuals of various age groups with iron, vitamin B_{12} and folic acid, but, they did not use supplements of vitamin A and vitamin C.

Imrana Khan (1992) carried out a study on anemic children, from low income groups in Karachi, to evaluate the effects of vitamin A supplementation on iron metabolism. This study did not include women of reproductive age group which is one of the most vulnerable groups.

Since the dietary intake of vitamins A, B, C, iron, folic acid, and proteins are all probably low in Pakistan in lower socio-economic strata, the women and children suffer from various degrees of mal-nutritional anemias and vitamin deficiencies.

The present study was therefore planned to establish weather vitamin A deficiency should or should not be a matter of concern in iron deficient anemic women in Pakistan.

When serum retinol was evaluated, it was seen that 6.2% of the women in the study group fell below the lowest reference limit.

Vitamin A is normally stored in liver (Passmore & Eastwood, 1986; Hoppner et al., 1968; Raica et al., 1972). Most of the values in all the subjects in the present study were on the lower side of normal if not sub-normal. This signifies that, probably, the subjects had no vitamin A stores at baseline, which is not unexpected, keeping in view the dietary patterns of this country, specially in the low/middle income groups.

It is also important to note that in the controls which were healthy women with adequate iron status, vitamin A status was found to be similar to that of the study group i.e. on the lower side of the reference limit.

More recently Anjum and Rana (1998) showed that vitamin A supplementation for 8 weeks failed to show significant improvement in hemoglobin status of anemic women of reproductive age group.

Keeping in mind the above studies and the results of the present survey, it can be suggested that:

1. Vitamin A deficiency is probably not the chief culprit in the etiology of iron deficiency anemia in Pakistani women.

2. A study involving greater number of subjects should be undertaken with intervention strategies.

3. The present study also indicates that the vitamin A status of Pakistani women is with in the normal range according to world standards, although lying on the lower side of the reference limit of the normal.

Table 1
Normal values[1] for the parameters investigated in the study (Harper, 1993)

Hemoglobin	12-16 g./dl.
Hematocrit	37-47%
RBC Count	4-5.5 x 10^6/ml.
Serum Iron	50-175 μg./dl.
Serum Ferritin	20-120 ng./ml.
Serum Transferrin	200-400mg./dl.
Serum Retinol	15-60 μg./dl.

[1]The values listed are for adult women.

Table 2
Mean values for the parameters investigated in controls and the study group

Group Title X 10^6/ml	Hemoglobin g/dl. mg/dl	Hematocrit % ng/ml	RBC count mg/dl	Serum iron mg/dl	Serum ferritin	Serum transferrin	Serum retinol
Controls (n=20)	11.04	31.96	4.14	66.48	28.91	166.20	32.60
Study Group (n=80)	8.68	30.64	3.74	53.0	18.10	132.52	29.67

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